Lesson notes · for you

Decisions: if and else

Programming · Level 2 · Lesson 3

Time	Materials
20 min	A phone with this lesson downloaded; Paper for a test table

Learning goals

- Compare with == and the other operators.
- Write if, elif and else with colons and indents.
- Join conditions with and and or.

Activity: The human ticket machine

1. Warm-up (3 min): Write the rule on paper: if age < 5, elif age < 12, else. Underline the colons and the indents.
2. Play (7 min): One pupil is the machine and may only follow the written rule. The others call out ages; the machine answers. Did anyone catch it out?
3. Test (7 min): In pairs, list the ages that are most likely to go wrong (4, 5, 11, 12) and write the ticket each should get.
4. Check (3 min): Run the program with each age from your table and tick the ones that match.

The machine follows only the rule on paper, not common sense. That is exactly how a computer works.

Discussion

- Why is 12 the age most likely to go wrong?
- When would you use and, and when or?
- Why does Python need two = signs to compare?

Check yourself

- Write an if line that is True only for ages 13 to 19.

Try it this week
Find a rule with an age or a price limit (a bus fare, a game's age rating). Write it as if, elif and else, and list the ages you would test.

Worksheet

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Name _____ Class _____ Date _____

1 **age = 15. Is this condition True or False?**

```
age > 12 and age < 18
```

◯ True ◯ False

2 **Finish the ticket machine: under 5 prints Free, under 12 prints Half price, everyone else Full price. It is tested at the boundaries.**

```
age = int(input("Age? "))
# under 5: Free, under 12: Half price, else: Full price
```

Write your program:

3 **A pupil can join the team if they are 13 or older and have a signed form. The program asks for the age first, then about the form. Put the lines in order. One line doesn't belong.**

Number the lines in order (1, 2, 3...), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `print("Welcome to the team!")`

☐ `if age >= 13 or form == "yes":`

☐ `print("Not yet")`

☐ `if age >= 13 and form == "yes":`

☐ `age = int(input("Age? "))`

☐ `else:`

☐ `form = input("Signed form? ")`

4 **Python stops this quiz with a SyntaxError. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  answer = input("2 + 2 = ? ")
2  if answer = "4":
3      print("Correct!")
4  else:
5      print("Try again")
```

5 **In if … elif … else, only one branch runs.**

◯ True ◯ False

✎ **Find a rule in your life with a limit (an age, a price, a time). Write it as if, elif and else. Which value would you test first?**

Check your answers

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1 **True**

15 > 12 is True and 15 < 18 is True. and needs both, and both are true, so the whole condition is True.

2

```
age = int(input("Age? "))
if age < 5:
    print("Free")
elif age < 12:
    print("Half price")
else:
    print("Full price")
```

The smallest band is tested first; elif catches the next; else gets everyone else.

3 **Top to bottom: 4, X, 6, 3, 1, 5, 2**

```
age = int(input("Age? "))
form = input("Signed form? ")
if age >= 13 and form == "yes":
    print("Welcome to the team!")
else:
    print("Not yet")
```

Both things must be true, so the condition uses and.

4 **Line 2: `if answer == "4":`**

Comparing needs ==. A single = means "store", and Python doesn't allow storing inside an if.

5 **True**

Python checks from the top and runs the first branch that is true, then skips the rest.